Task 3 :- Virtualization Components

Hypervisor (The Building Manager) :-

The Hypervisor is core technology behind virtualization. It's the software that creates and manages lab machines.

It is special piece of software that:

* Divides physical computer into multiple virtual ones.
* Gives each lab machine its own share of CPU, memory and storage.
* Keeps everything isolated and safe.
* Manages the lifecycle of lab machines (Start, Stop, pause, clone, delete).

Hypervisors have 2 main types of implementation, each used for specific scenarios, from home labs to large data centers.

Type 1 :- hypervisors run directly on physical hardware, making them fast, efficient and ideal for servers and professional environments.

Type 2 :- hypervisors run within existing operating system, making them easier to install and ideal for learning, testing or small setups.

use case	Type 1	Type 2
Test Malicious file		X
production Server	X	
Database Server	X	
Software Testing		X
Kali Linux		X
Data center	X	

When using virtualization to test malicious files, care should be taken to ensure that host machine does not become infected by malware being tested in guest machine.

One approach is to use different operating systems for guest and host machines, or to isolate guest machines so that it does not communicate with the host.

Lab Machines (The Apartments):—

* A lab Machine (VM) is virtual computer created by hypervisor.

* Even though its virtual, it behaves as real machine:

→ It has its own CPU, RAM, storage and network.

→ It can run on any operating system (windows, linux etc).

→ Its completely isolated from other VMs. This means that if one VM breaks, the other continue to work.

Ex:- You can deploy VMs on your own computer using tools such as Oracle virtual box and VMware workstation. This type of software acts as type 2 hypervisor and lets you run multiple operating systems, such as windows, linux and Mac OS.

Lets see some examples since you have learned hypervisor and VM.

* You need to work on different OS like kali linux, but you can't buy another whole system, so install hypervisor and run kali linux VM on it.

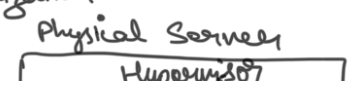
* you want to test whether a file is malicious, so you set up isolated lab machine to protect your main computer from being infected.

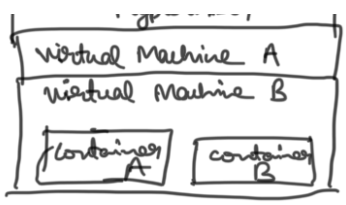
Containers (The Rooms inside Apartment):—

A Container is lightweight isolated Environment that runs a single application and all necessary components to support it. Instead of bringing whole separate operating system, a container borrows core of existing system by running on kernel, which is part of operating system that communicates with hardware and manages resources such as memory and running programs. Because containers share this kernel, they start quickly, and use few resources than full lab machines, Example you can't run windows container on linux machine.

The Easiest way to deploy Containers in VM is using Docker.

* Docker is open source software platform that simplifies process of building, deploying and running applications using containerization.





In Summary containers provide "full Apartment" with maximum seperation and flexibility, while containers offers lightweight "rooms" Ideal for Scalable, fast deploying applications.

Quiz:-

1) Suppose a user wants to deploy Study lab on there machine to practice some exercises for cybersecurity certification. which type of hypervisor will they use?

A) Type 2

2) Suppose a company wants to host multiple small applications in same lab machines. what should they use?

A) Containers.